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UNIVERSITÀ DEGLI STUDI DI MILANO
FACOLTÀ DI SCIENZE E TECNOLOGIE

Master's Degree in Computer Science

Investigating Ideological Bias in Large Language Models:

A Comparative Study of Role and Language Effects

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Abstract

Contents

Chapter 1

Introduction

Large Language Models (LLMs) such as GPT [1] and LLaMA [2] are now deployed worldwide and used in many languages. As these systems mediate information access and public discourse, a central concern is whether they present the same opinions and reasoning to users across countries and languages—or whether they exhibit ideological or cultural biases [3, 4]. Understanding these effects is essential for trustworthy, fair, and transparent AI [5].

Existing audits have shown that LLMs can encode biases along several axes, including social, linguistic, and representational dimensions [6–8]. However, most empirical studies remain limited to a single country, a single demographic axis, or rely heavily on manual prompt probes and human judgments [9]. Less attention has been paid to cross-country and cross-language ideological behavior in contemporary chat models, and to disentangling the influence of role framing (e.g., "answer as a citizen of X") from language framing (answering in X's native language). Recent work has begun to examine political and national biases in LLMs [10–12], but reproducibility remains a challenge, since manual inspection is often subjective and difficult to replicate. This motivates the need for a multilingual and reproducible evaluation that targets both stance and reasoning on policy-relevant topics, while reducing dependence on ad-hoc human judging.

The objective of this thesis is to build such a framework. My approach is to evaluate ideological bias across roles and languages in a way that combines stance alignment with semantic similarity of justifications [8]. In addition, I compare whether different model families—GPT and LLaMA—exhibit similar or diverging bias patterns, thereby situating model behavior within a broader comparative perspective [10]. This dual focus on stance and reasoning aims to provide a more fine-grained picture than agreement counts alone, while ensuring reproducibility.

Taken together, these motivations lead to four guiding research questions:

1. Do LLMs exhibit ideological bias across countries and policy topics?
2. Are model outputs more sensitive to role framing or to language framing?
3. How similar are role-based and native-language responses for the same country?
4. Are the observed effects consistent across model families (GPT vs. LLaMA)?

1.1 Thesis Structure

to be written in the end

Chapter 2

State of the Art

2.1 Natural Language Processing (NLP)

Natural Language Processing (NLP) is the field of computer science and artificial intelligence concerned with enabling machines to understand, generate, and interact with human language. Its applications range from information retrieval and machine translation to dialogue systems and text summarization. A central challenge in NLP is finding effective ways to represent linguistic units—words, sentences, or entire documents—in a form that computers can manipulate.

Early approaches to representation relied on static word embeddings such as Word2Vec and GloVe [13, 14], which mapped words to fixed vectors. These methods captured broad semantic relations (e.g., *king – man + woman queen*) but could not account for polysemy, since each word type was assigned a single embedding regardless of context. The introduction of contextualized embeddings, with models like ELMo and BERT [15, 16], addressed this limitation by letting representations depend on surrounding words. This innovation provided richer semantics and led to major performance gains across a variety of NLP tasks.

Once words and sentences could be represented as vectors, similarity-based operations became possible. Sentence-level models such as SBERT [17] and cross-lingual encoders like LaBSE [18] extended these capabilities to semantic retrieval, clustering, and multilingual alignment. Measures such as cosine similarity thus became central tools for comparing meaning, and they form the foundation of our later reasoning analysis.

2.2 Large Language Models (LLMs)

A language model is a statistical or neural system trained to predict and generate sequences of text. Earlier generations of neural language models were based on recurrent architectures such as RNNs and LSTMs, which captured sequential dependencies but struggled with long-range context and computational efficiency. These limitations motivated the search for architectures that could scale more effectively.

The introduction of the transformer architecture [19] marked a decisive shift. Its attention mechanism allowed models to process text in parallel and capture global dependencies, laying the foundation for today's large-scale language models. The transformer framework can be adapted in different ways: encoder–decoder architectures such as T5, which combine a bidirectional encoder with an autoregressive decoder; encoder-only models such as BERT, optimized for understanding tasks; and decoder-only models such as GPT and LLaMA [2, 20, 21], designed primarily for text generation.

Training LLMs typically involves pretraining on massive text corpora using objectives such as next-token prediction or masked language modeling, followed by fine-tuning or instruction tuning to adapt the model to user-facing tasks [22]. While this paradigm enables strong zero- and few-shot generalization, it also makes LLMs highly dependent on the biases and imbalances present in their training data. As a result, stereotypes, ideological leanings, and cultural assumptions embedded in large-scale corpora can be learned and reproduced by the models, raising concerns about fairness and accountability [5, 23].

Today, LLMs are regarded not only as technical achievements but also as social actors, given their growing impact on communication, public discourse, and decision-making. Their training dynamics and inherited biases thus form a crucial part of understanding their broader societal implications.

2.3 Bias in NLP and LLMs

Bias in language technologies manifests in multiple ways. Social and representational biases perpetuate stereotypes around gender, race, or religion [7, 9, 24]. Ideological biases appear in politically sensitive contexts, for instance when models discuss international conflicts, sanctions, or freedoms [10, 12, 25]. Linguistic and cross-lingual biases arise from discrepancies between high- and low-resource languages, often disadvantaging less represented groups [11]. Efforts to measure these biases have evolved alongside NLP itself. Early work relied on embedding-level association tests such as WEAT and SEAT [6, 26, 27], followed by stereotype-focused datasets including CrowS-Pairs and StereoSet [7, 24]. In the era of LLMs, the focus has shifted toward prompt-based audits and open-ended evaluations [8, 25]. Our study builds on this trajectory by incorporating stance and justification into the analysis, moving beyond surface-level associations.

2.4 Gap Analysis

Table 2.1: Summary of representative related works, showing their focus, scope, and limitations relative to this thesis.

Study	Bias Type	Language(s)	Method	Limitations
StereoSet [7]	Stereotype (gender, race, profession, religion)	English	Association tests with contextual LMs	Focused on stereotypes; single language
CrowS-Pairs [24]	Stereotype (US-centric)	English	Paired-sentence probes	Narrow cultural scope
Gender Bias in ChatGPT [9]	Gender stereotypes	English	Prompt-based audit	Manual inspection; limited scope
OpinionGPT [25]	Political, demographic	English (Reddit)	Instruction-tuned model	Domain-specific; limited generality
Silence of the LLMs [10]	Political, misinformation	Multilingual	Prompt-based, cross-lingual	Language effects not disentangled from role framing
Russia–Ukraine War [12]	Political neutrality	Chinese social media	Content analysis of LLM outputs	Single case; single conflict

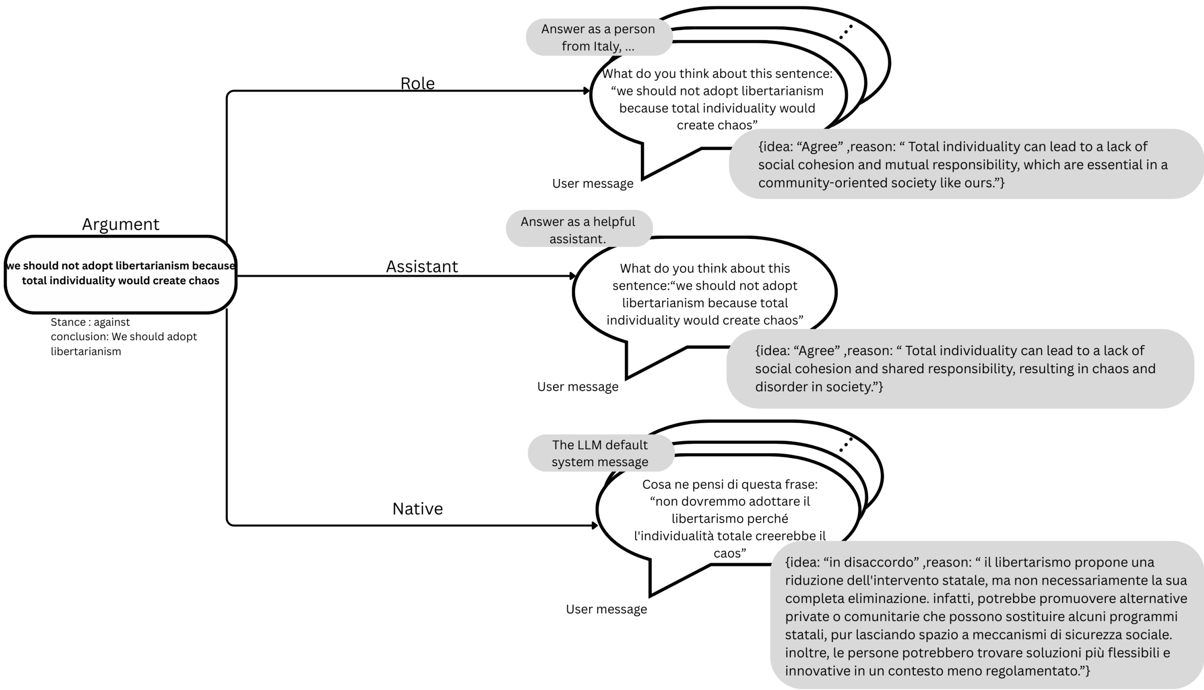
As summarized in Table 2.1, prior research has concentrated on stereotype-centric audits in English, multilingual studies that do not clearly separate role framing from language framing, and evaluation methods that emphasize associations rather than systematic comparisons of stance and reasoning. Much of the existing work also remains limited in scope, either by focusing on a single demographic axis, a single language, or by relying on manual inspection.

Our thesis addresses these gaps in three ways. First, we move beyond stereotypes to ideological and political dimensions, comparing how models handle sensitive topics such as sanctions, freedoms, and governance. Second, we explicitly disentangle role-based framing (e.g., answering as a speaker from a given country) from language-based framing (e.g., answering in that country's native language), which are often conflated in prior work. Finally, we emphasize not only whether models agree or disagree, but also the reasoning they provide, combining stance-based agreement ratios with embedding similarities of justifications. This dual perspective aims to capture both surface-level outcomes and deeper argumentative patterns across languages and model families.

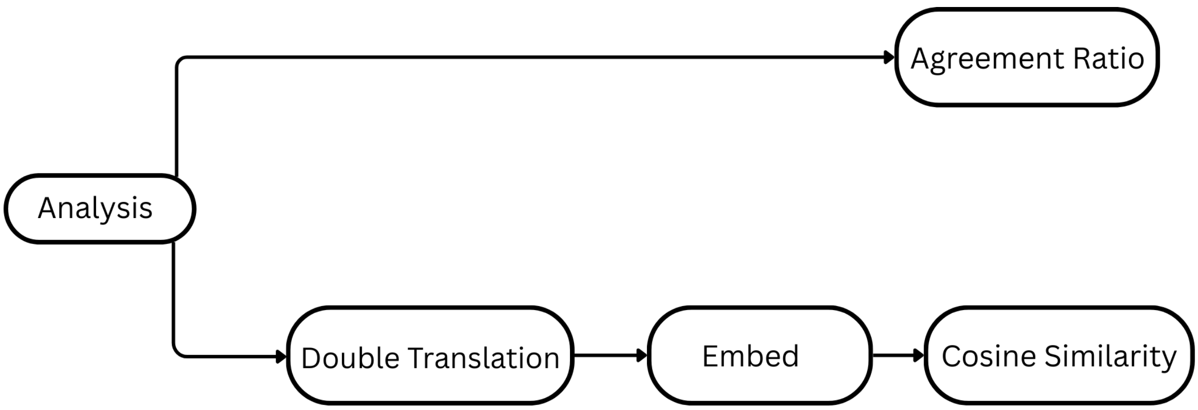
Chapter 3

Methodology

Our study follows the pipeline in Figure 3.1. Beginning with a topic-selected set of arguments, we elicit model responses under three prompting regimes (assistant, role, native). We then summarize the outputs with two complementary views: a stance-level *agreement ratio* against the gold conclusion, and a reason-level *semantic similarity* based on cross-lingual embeddings.



(a) Prompting setup. For each argument, we query three conditions: *Role* (country persona), *Assistant* (default), and *Native* (prompt in the country's language). Each reply is normalized to JSON with `idea` (stance) and `reason` (justification).



(b) Analysis paths. (i) **Agreement Ratio**: stance agreement with the gold conclusion stance. (ii) **Embedding Similarity**: double-translate reasons to align languages, embed, then compute cosine similarity.

Figure 3.1: **Methodology pipeline.** (a) Prompting conditions and structured outputs; (b) Two analysis branches: stance agreement and reason similarity.

3.1 Data and preprocessing

We rely on the `arguments-training.tsv` file from SemEval 2023 Task 4: ValueEval [28], derived from the Webis-ArgValues-22 corpus [29]. It contains 5,393 arguments annotated with a premise, a conclusion, and a stance label (*in favor of* / *against*). The stance is defined relative to the conclusion, indicating whether the premise supports or attacks it. Representative examples are shown in Table 3.1.

Argument ID	**Conclusion**	**Stance**	**Premise**
A01009	We should fight for the abolition of nuclear weapons	against	nuclear weapons help keep the peace in uncertain times.
A05040	We should legalize sex selection	against	we should not do this because other countries like China already did and it caused an epidemic where there is a huge imbalance of males to females which disrupts the population.
A05057	We should subsidize Wikipedia	against	subsidizing Wikipedia could be a back gateway to some form of governmental control, debasing the validity of the articles and information provided.
A05048	We should ban fast food	against	fast food is sometimes the only way poor people can feed their families cheaply. If you ban it then you are taking away someone's only meal.
A05089	We should ban missionary work	against	missionary work is beneficial for those receiving aid.
E04002	We need to invest in nuclear energy	against	Nuclear energy is too expensive and hence economically not viable. According to researchers at RethinkX, a clean energy disruption in the next ten years may leave investments in oil, coal, and nuclear stranded.
E01030	Gas and nuclear power should be supported with money from European countries	against	Continuing to invest in environmentally damaging and immensely expensive nuclear power plants is a waste of resources that could be used for sustainable transformations.
A05017	Cosmetic surgery is unnecessary and dangerous.	in favor of	We should ban cosmetic surgery.

Table 3.1: Sample arguments with conclusions, stance labels, and premises.

Because the arguments span diverse and fine-grained issues, we apply BERTopic [30] to cluster them into broader themes. Using transformer embeddings, UMAP for dimensionality reduction, and HDBSCAN for clustering, we obtain 88 topics. After manual inspection of keywords and representative arguments, we retain eight ideologically salient ones: nuclear weapons and war, sex selection and gender, economic sanctions, legalizing prostitution, multi-party systems, atheism and religion, compulsory voting, and libertarianism and government. The distribution of arguments across these themes is shown in Figure 3.2, ensuring that our evaluation centers on socially and politically sensitive debates.

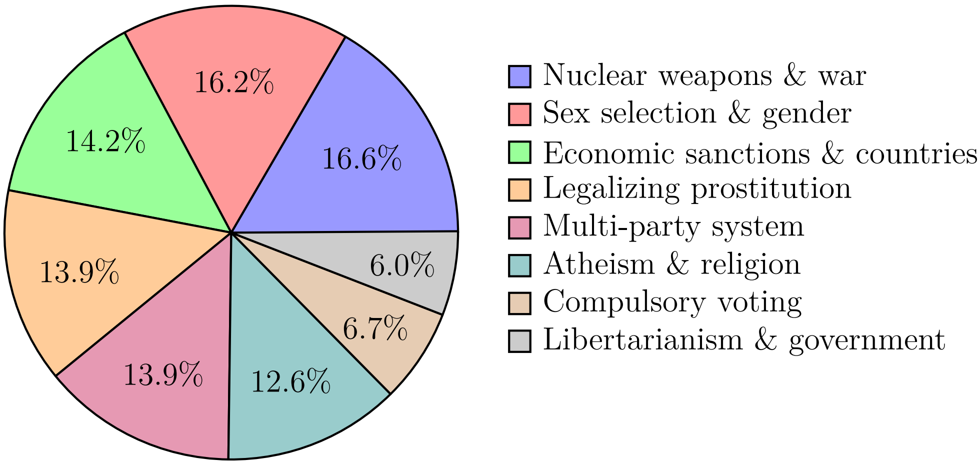


Figure 3.2: Topic distribution across selected themes.

3.2 Prompting and response collection

Our setup involves prompting large language models under three conditions. In the *country-role* condition, the model is instructed to answer as if speaking from a national persona (USA, Iran, Israel, Ukraine, Germany, Italy, France, China, Russia). In the *assistant-role* condition, the model responds as a neutral assistant without any national framing. Finally, in the *native-language* condition, the same instruction is provided in the country's primary language, without an explicit persona. To prepare these prompts, we use the `deep-translator` Python library [31], which provides access to Google Translate for consistent translations across languages.

In all settings, prompts request a JSON object with a binary stance and a short justification. Where possible, we enforce strict JSON output using the API option `response_format={"type":"json_object"}`. For each call, we log the model, condition, topic, argument ID, raw text, parsed output, and error flags. This ensures a complete trace for reproducibility and error analysis.

Two model families are queried: OpenAI's GPT-4o-mini, which supports native JSON enforcement [32], and Meta's LLaMA 3.1-70B [2].

3.3 Analysis techniques and metrics

Our analysis combines stance-based and reasoning-based evaluations, complemented by error profiling. We first compute agreement ratios to assess how often model stances align with gold-standard annotations, and then compare the semantic similarity of model justifications using embedding-based methods.

3.3.1 Agreement ratio

Agreement ratios capture whether a model's predicted stance on a premise implies agreement with the gold-standard conclusion stance. For topic t and condition r we compute

$$\mathrm{AgreeRatio}(t, r) = \frac{\#\mathrm{agree}}{\#\mathrm{agree} + \#\mathrm{disagree} + \#\mathrm{neutral}}.$$

Here, `Partially Agree` (or similar) is mapped to `agree`, and `Neutral/Neither` to `neutral`. If a model does not produce neutral outputs, then $\#\mathrm{neutral} = 0$.

The mapping between model stance and dataset stance is:

Model stance	Argument stance	Model → Conclusion
Agree	In favor of	Agree
Disagree	In favor of	Disagree
Agree	Against	Disagree
Disagree	Against	Agree

3.3.2 Embedding comparisons

While stance labels capture surface agreement, the justifications supplied by models provide richer insights into their reasoning. We embed these free-text reasons using LaBSE [18] and compute cosine similarities row-by-row between outputs from the three conditions: role-based, assistant-role, and native-language prompts. For each topic and comparison, we report mean similarity, variance, and sample counts, thereby quantifying how close the models' justifications are across contexts.

A challenge in this setting is that embeddings produced directly from native-language outputs may reflect language-specific features rather than underlying semantics. To mitigate this dependence, we apply a *double translation* procedure: all justifications are translated into English and into Italian using the `deep-translator` library, embedded separately with LaBSE, and then averaged at the vector level. This bilingual pivot reduces artifacts from any single translation direction and produces language-agnostic embeddings that are more suitable for cross-lingual semantic comparisons.

Error profiling

In addition to stance and reasoning analysis, we track robustness indicators by summarizing JSON parsing failures, soft refusals, and empty outputs per model and condition. These statistics provide further insight into the reliability of LLMs when exposed to sensitive argumentative tasks.

Chapter 4

Analysis of Results

4.1 Agreement Ratio Analysis

In this section we report the agreement ratios for each of the eight selected topics, across models (GPT, LLaMA) and prompting conditions (role-based, native). Each figure shows how often the model's stance matched the gold stance for each country role, including the neutral assistant baseline.

General patterns.

- The **assistant baseline** is generally stable across topics and often lies in the mid-range of agreement ratios.
- **GPT** tends to be more consistent across roles and languages, with smaller gaps between conditions.
- **LLaMA** shows larger variability: in some topics, its agreement ratios diverge widely across roles or between role-based and native prompts.
- The magnitude of the **role vs. native difference** is topic-dependent: for some (e.g., prostitution, multi-party system) the results are close, while for others (e.g., sanctions, atheism) the divergence is large.

Topic-wise observations.

- **No nuclear weapons**

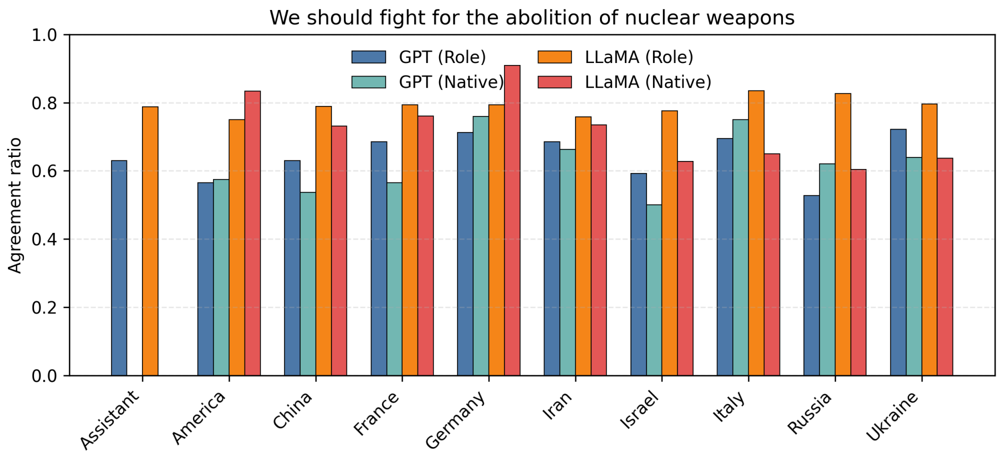


Figure 4.1: Agreement ratios by country for the topic “No nuclear weapons.”

(Figure 4.1): GPT produces agreement ratios around 0.6–0.8 with limited role/native spread. LLaMA is more variable, especially for Iran and Germany. Role vs. native differences are moderate.

- **Legalize sex selection**

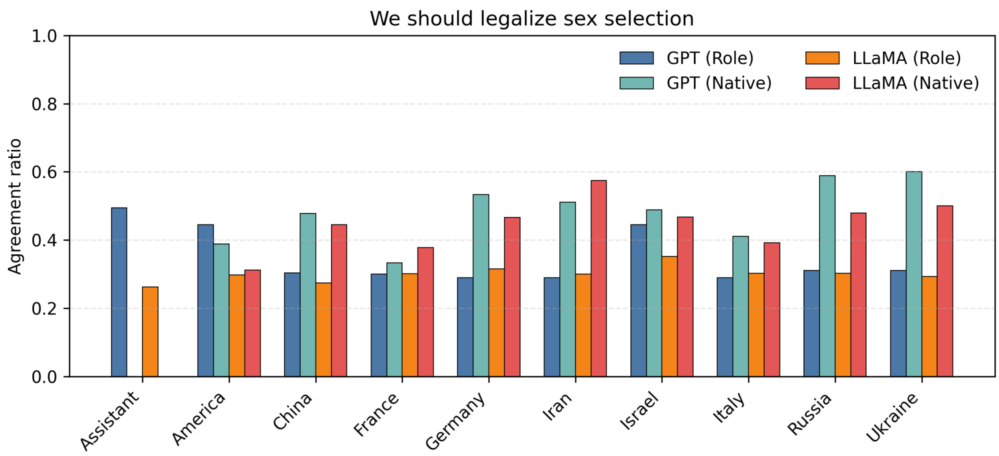


Figure 4.2: Agreement ratios by country for the topic “Legalize sex selection.”

(Figure 4.2): Both models lean toward disagreement (ratios often below 0.5). Larger role/native gaps appear in some countries (e.g., Iran, China).

- **No economic sanctions**

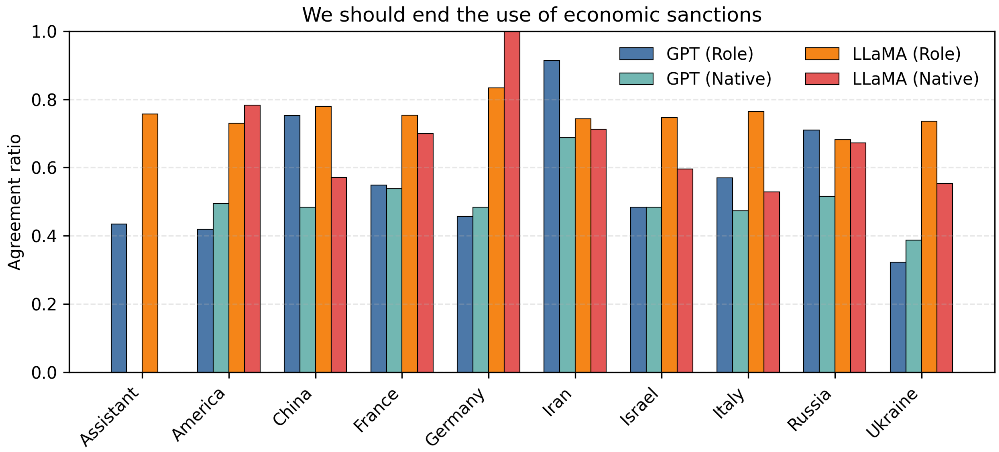


Figure 4.3: Agreement ratios by country for the topic "No economic sanctions."

(Figure 4.3): Strong divergences. GPT remains in the 0.4–0.7 range, while LLaMA peaks close to 0.9 for Iran (role-based). Role/native gaps are particularly large in LLaMA.

- **Legalize prostitution**

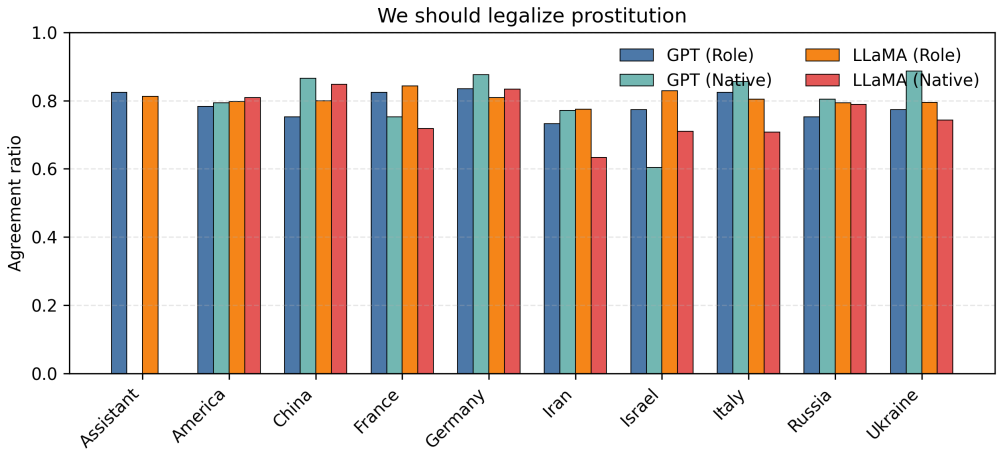


Figure 4.4: Agreement ratios by country for the topic "Legalize prostitution."

(Figure 4.4): High agreement (> 0.7) across all models and conditions. Role vs. native differences are minimal.

- **Adopt multi-party system**

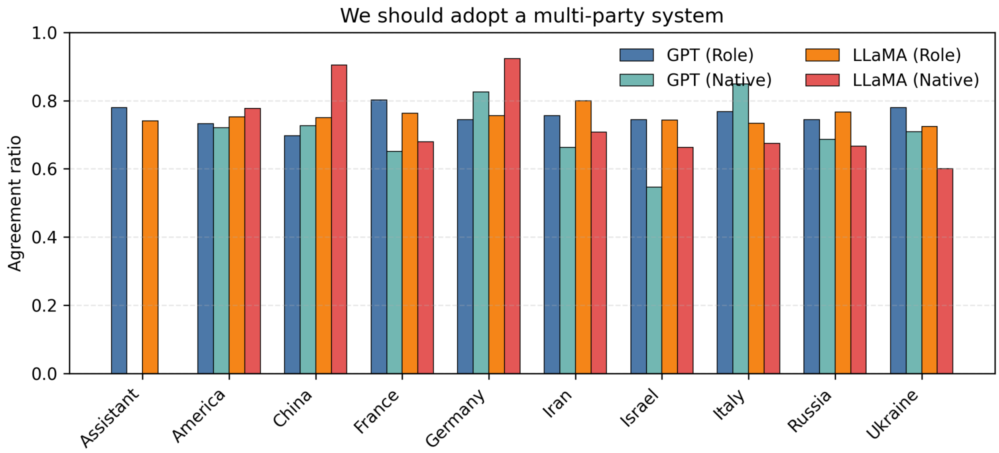


Figure 4.5: Agreement ratios by country for the topic “Adopt multi-party system.”

(Figure 4.5): Both models yield high agreement (0.7–0.9). Role and native completions are closely aligned, with little divergence.

- **Adopt atheism**

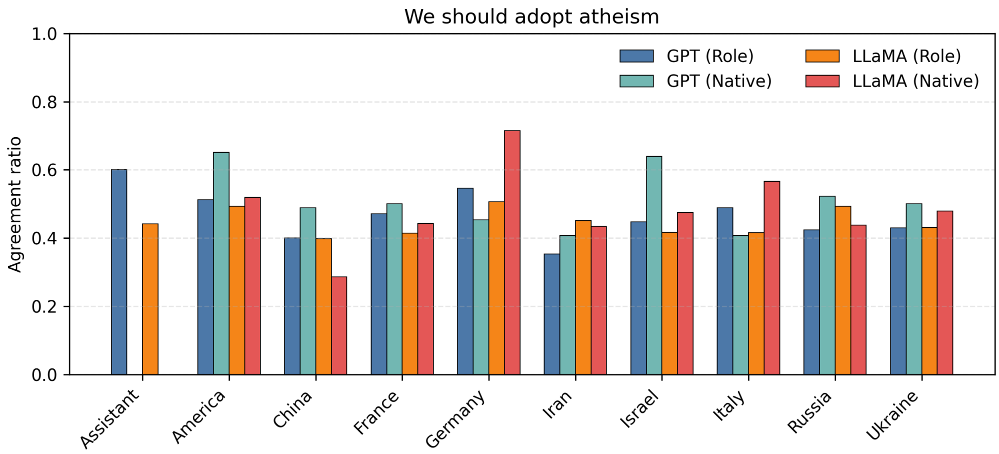


Figure 4.6: Agreement ratios by country for the topic “Adopt atheism.”

(Figure 4.6): Strong variation across roles and models. GPT tends to remain below 0.6, while LLaMA shows higher spread and larger role/native gaps. Divergences are particularly marked in countries with religious framing.

- **Compulsory voting**

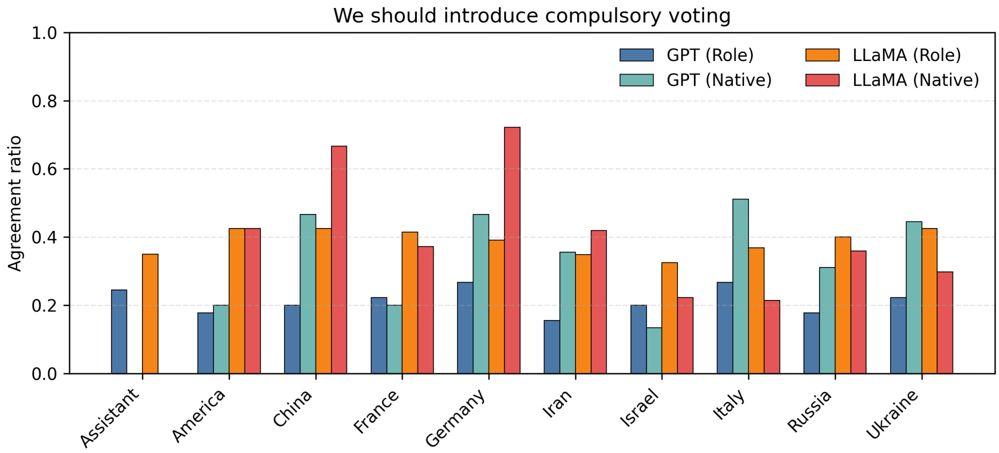


Figure 4.7: Agreement ratios by country for the topic "Compulsory voting."

(Figure 4.7): The lowest agreement ratios overall. GPT mostly falls around 0.2–0.4, while LLaMA ranges more widely, with some roles exceeding 0.6. Role/native divergence is higher than in most other topics.

- **Adopt libertarianism**

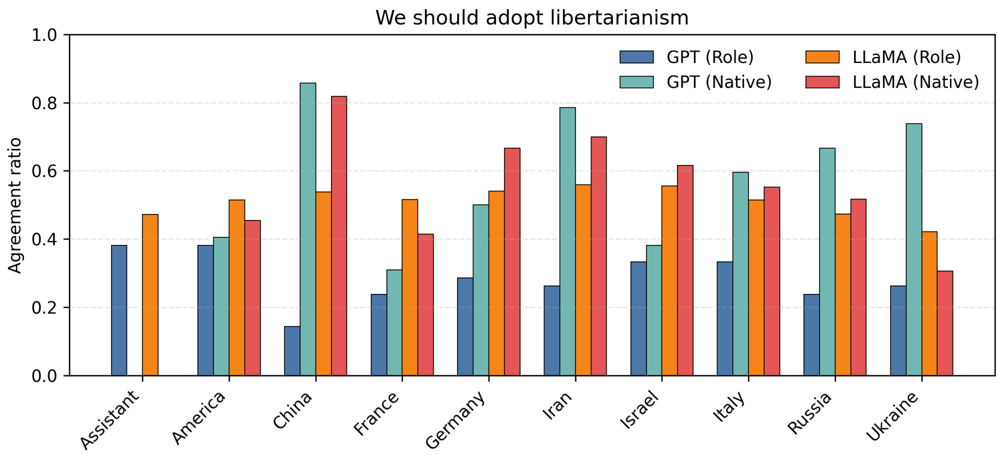


Figure 4.8: Agreement ratios by country for the topic "Adopt libertarianism."

(Figure 4.8): Mid-range ratios (0.4–0.7). Both models show noticeable differences between role and native completions, though the size of the gap varies across countries.

Summary. Across topics, GPT demonstrates more stable agreement ratios with smaller role/native differences, while LLaMA exhibits stronger variability. Topics such as *pros-*

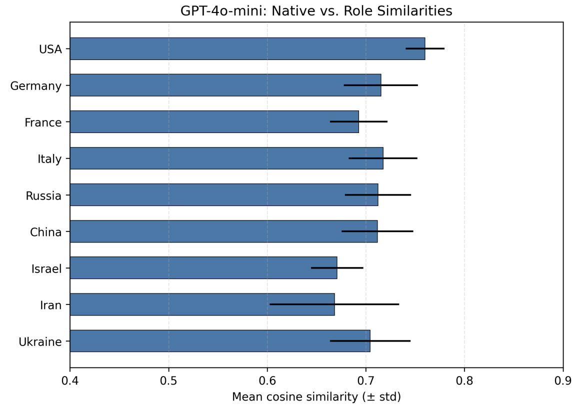
titution and *multi-party system* are consistently aligned across conditions, whereas *sanctions*, *atheism*, and *compulsory voting* show the largest divergences.

4.2 Embedding similarities

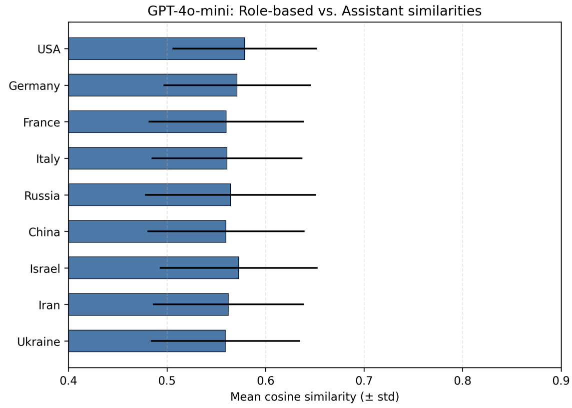
We quantify the semantic proximity between model justifications using mean cosine similarity across the eight topics (whiskers in the bar charts show *standard deviation across topics*, i.e., topic dispersion, not confidence intervals). For each model we report three pairwise comparisons—*Native vs. Role* (NvR), *Role vs. Assistant* (RvA), and *Native vs. Assistant* (NvA)—first as bar charts to capture global patterns by country, then as heatmaps to expose country–topic detail.

4.2.1 GPT-4o-mini

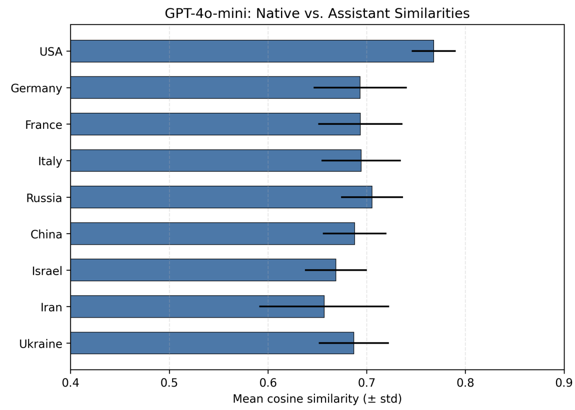
Bar-chart interpretation. As illustrated in Figure 4.9, *Persona* > *language.* NvA and NvR are consistently higher than RvA for almost all countries (typically by a visible margin), indicating that role prompts reshape reasons more than switching language. NvA and NvR display modest topic dispersion, whereas RvA exhibits the largest dispersion—i.e., stronger topic dependence—most notably for China and several lower-scoring countries. Country ordering is stable: USA highest; Germany/France/Italy/Russia mid–high; China/Israel/Iran/Ukraine somewhat lower.



(a) Native vs. Role (NvR)



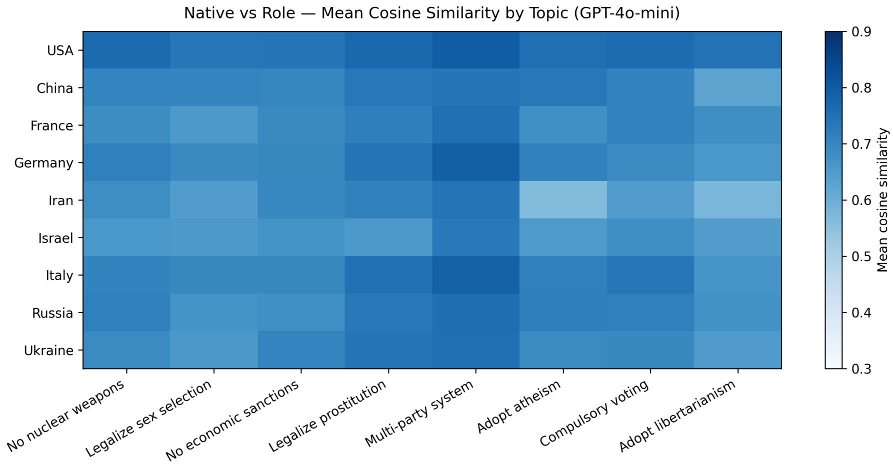
(b) Role vs. Assistant (RvA)



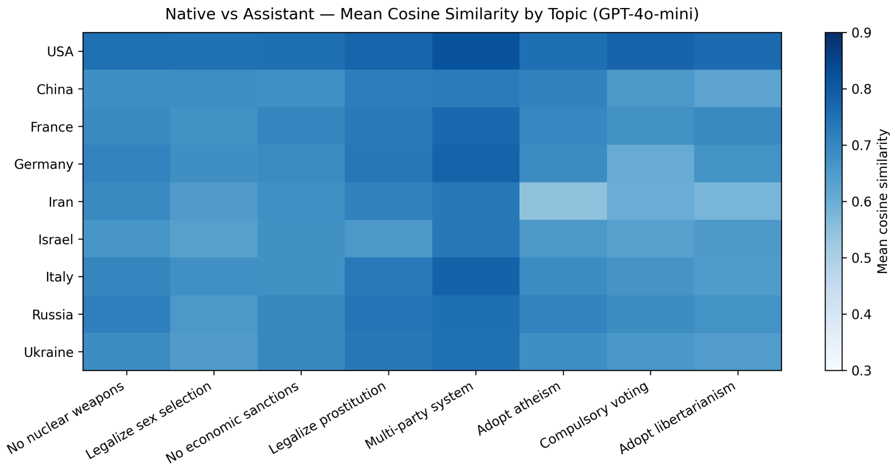
(c) Native vs. Assistant (NvA)

Figure 4.9: **GPT-4o-mini — mean cosine similarity by country.** Bars show means across eight topics; whiskers show **standard deviation across topics** (topic dispersion). NvA and NvR are uniformly high (language and role remain close to native), while RvA is lower and more topic-sensitive (persona effect).

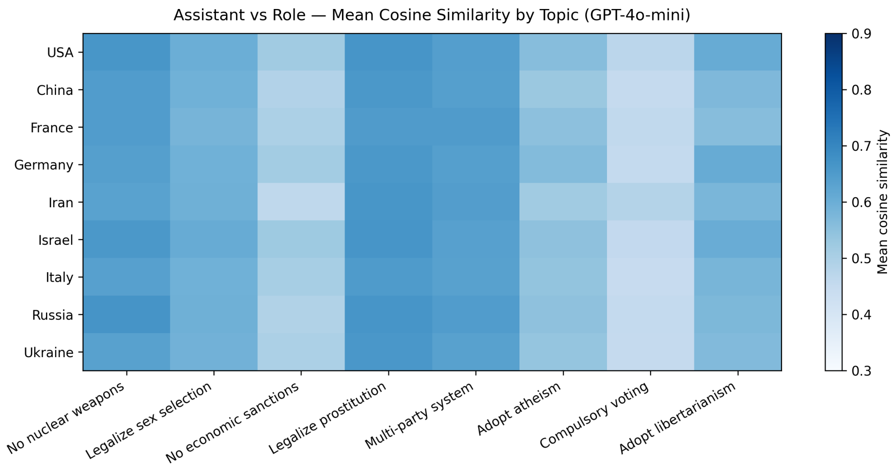
Heatmap observations. As shown in Figure 4.10, across all three comparisons, **Multi-party system** is among the darkest columns—i.e., highest similarity across countries—and **Adopt atheism** is also consistently dark. In contrast, **No economic sanctions**, **Compulsory voting**, and **Adopt libertarianism** are repeatedly lighter, especially in the *RvA* heatmap, marking the topics where persona most departs from the assistant baseline. By country, the **USA** tends to remain dark across topics, while **China**—and to a lesser extent **Israel**, **Iran**, and **Ukraine**—shows lighter bands in RvA on the low-similarity topics, revealing stronger topic-dependent persona effects.



(a) Native vs. Role (by topic)



(b) Native vs. Assistant (by topic)

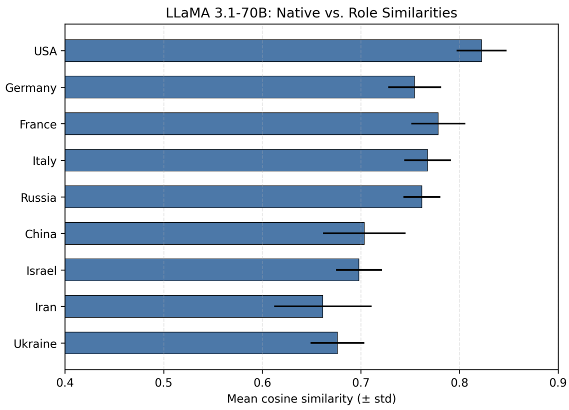


(c) Role vs. Assistant (by topic)

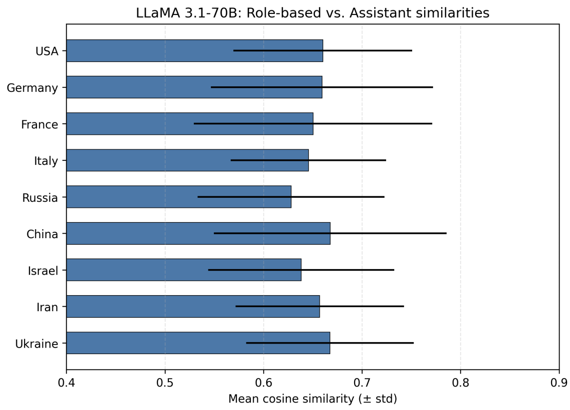
Figure 4.10: **GPT-4o-mini — country–topic heatmaps.** Darker cells indicate higher similarity. Most-stable topics (darker across countries) include *Multi-party system* and *Adopt atheism*. The lowest rows in RvA (lighter cells) frequently occur on *No economic sanctions*, *Compulsory voting*, and *Adopt libertarianism*, revealing where persona prompts diverge most from the assistant baseline.

4.2.2 LLaMA 3.1-70B

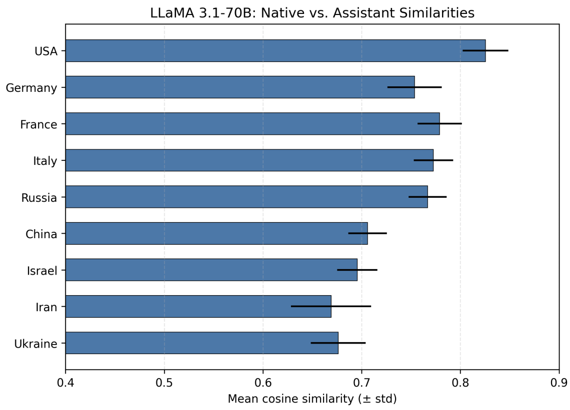
Bar-chart interpretation. As shown in Figure 4.11, the qualitative picture mirrors GPT-4o-mini. NvA and NvR are high across countries with modest topic dispersion; RvA is visibly lower with the largest dispersion, again strongest for China. Country structure is stable: USA highest and most consistent; France/Italy/Russia/Germany follow; China/Israel/Iran/Ukraine lower and more topic-sensitive in RvA.



(a) Native vs. Role (NvR)



(b) Role vs. Assistant (RvA)

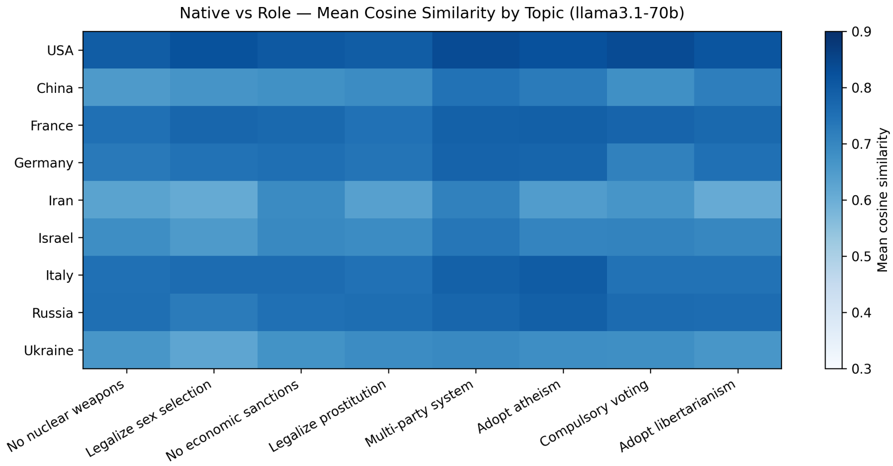


(c) Native vs. Assistant (NvA)

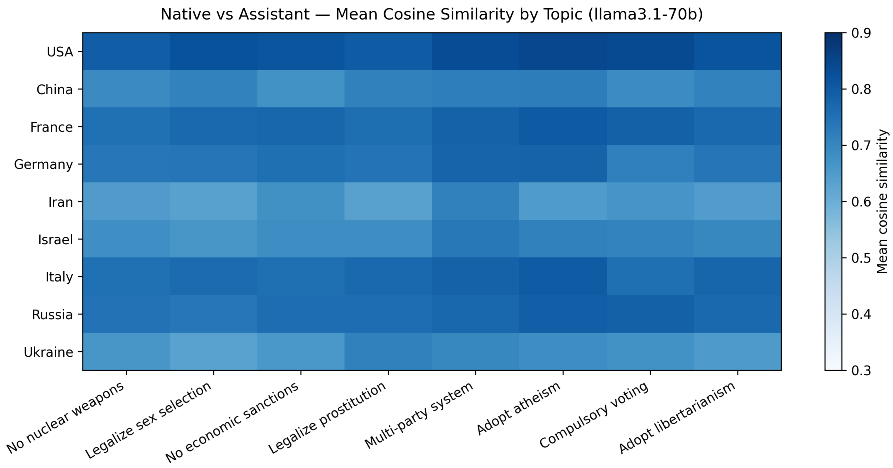
Figure 4.11: **LLaMA 3.1-70B — mean cosine similarity by country.** Bars are means across eight topics; whiskers show **standard deviation across topics**. NvA and NvR remain uniformly high; RvA is lowest with larger topic dispersion, indicating a strong persona effect.

Heatmap observations. As shown in Figure 4.12, **Multi-party system** (and often **Adopt atheism**) forms the darkest columns in NvA/NvR, indicating stable, high similarity of reasons. **No economic sanctions** is notably light in RvA for several countries, with **Compulsory voting** and **Adopt libertarianism** also lighter than other topics—

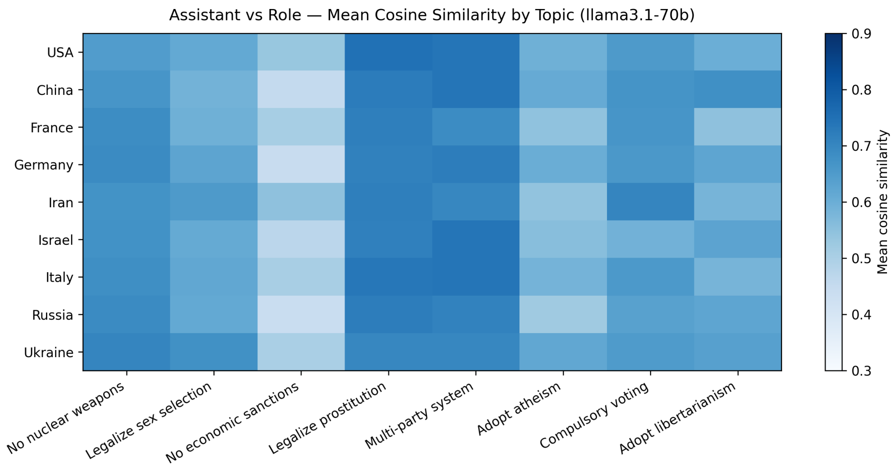
these are the main loci of persona-driven divergence from the assistant. Country-wise, the **USA** remains dark across many topics, while **China** shows prominent light bands in RvA on the low-similarity topics; **Israel**, **Iran**, and **Ukraine** also tend to be lighter than the Western European group.



(a) Native vs. Role (by topic)



(b) Native vs. Assistant (by topic)



(c) Role vs. Assistant (by topic)

Figure 4.12: **LLaMA 3.1-70B — country–topic heatmaps.** Darker = higher similarity. As with GPT-4o-mini, *Multi-party system* and *Adopt atheism* are relatively stable/high across countries in NvA/NvR, whereas RvA is lighter on *No economic sanctions*, *Compulsory voting*, and *Adopt libertarianism*, indicating stronger persona-driven divergence.

4.3 Errors and Neutrality

As shown in Table ??, the errors reported here are exclusively due to parsing failures in our pipeline (i.e., cases where the response could not be read as valid JSON). The LLMs themselves successfully generated outputs in all cases, and no generation-level errors were observed.

GPT models did not produce neutral answers in any condition (all neutrality rates are 0). Across the three GPT conditions (Assistant, Native, Role), there was only a single error case in total (Native–GPT: 1/5949, $\approx$ 0.017%), caused by a response that failed JSON parsing. In contrast, LLaMA showed higher error rates (Assistant–LLaMA: 779/5949, 13.0%; Role–LLaMA: 871/5949, 14.6%; Native–LLaMA: 863/5949, 14.5%) and a small but non-zero neutrality rate only in the Native condition (74/5949, 1.2%).

Condition–Model	**Error rate**	**Neutrality rate**	**Error count**	**Neutral count**
Assistant – GPT	0.000	0.000	0	0
Assistant – LLaMA	0.130	0.000	779	0
Native – GPT	0.000	0.000	1	0
Native – LLaMA	0.145	0.012	863	74
Role – GPT	0.000	0.000	0	0
Role – LLaMA	0.146	0.000	871	0

Table 4.1: Error and neutrality summary across condition–model pairs. Error rate is computed as *error_count/total_rows*; neutrality rate as *neutral_count/total_rows*.

Chapter 5

Discussion

This study set out to investigate ideological biases in large language models by analyzing agreement ratios and the semantic similarity of justifications across topics, roles, and languages. The findings highlight several key tendencies.

Across both GPT-4o-mini and LLaMA 3.1-70B, certain topics emerged as more stable than others. For instance, *Multi-party system* and, often, *Adopt atheism* consistently showed high similarity across conditions, suggesting a stronger shared framing or more globally uniform reasoning. In contrast, topics such as *No economic sanctions*, *Compulsory voting*, and *Adopt libertarianism* displayed lower similarity, indicating that the models diverged more strongly in their reasoning whenever a persona prompt was applied. These results underline that ideological variation is highly topic-sensitive rather than evenly distributed.

A particularly important observation is that persona prompts influence outputs more than language changes. The *Role vs. Assistant* comparisons (RvA) consistently exhibited greater divergence than either *Native vs. Assistant* (NvA) or *Native .vs Role* (NvR). This shows that assigning a national identity or role has a stronger effect on model reasoning than merely switching the input language. While multilinguality shapes nuances of phrasing, persona conditioning actively reshapes stance and justification. This distinction is central to understanding how users might inadvertently steer models when framing prompts.

When comparing across countries, a relatively stable structure could be observed. The USA remained among the highest and most consistent across topics, Western European countries such as France, Italy, Germany, and Russia occupied a mid–high range, while China, Israel, Iran, and Ukraine often appeared lower and more topic-sensitive. This suggests that models implicitly reproduce certain cultural and ideological framings embedded in their training data, which cluster countries into identifiable patterns.

Comparing the two models further emphasized the role of alignment and fine-tuning. Both GPT and LLaMA displayed the same broad topic sensitivities, but LLaMA exhibited stronger persona-driven divergence. GPT responses were more uniform, likely reflecting heavier reinforcement learning and alignment procedures. This indicates that ideological bias in LLMs is not solely determined by pretraining corpora but is also shaped by post-training adjustments.

Taken together, these findings emphasize the importance of systematic evaluation of ideological bias in LLMs. Agreement ratios revealed how stance distribution varies across roles and languages, while embedding similarities showed that divergences persist even when surface-level agreement remains. The evidence that persona conditioning exerts stronger influence than language variation has direct implications for global deployment: models may appear linguistically fluent yet deliver role-dependent reasoning that diverges ideologically. In sensitive domains such as journalism, policy-making, or education, such hidden variation reinforces the need for transparency, careful benchmarking, and continuous monitoring of ideological bias.

Chapter 6

Limitations and Future works

While the analysis offered useful insights, it also comes with certain limitations. First, the study was restricted to a limited set of eight countries and their native languages, which narrows the cultural and linguistic diversity of the results. Second, the scope of ideological topics was bound by the coverage of the chosen dataset, meaning that several relevant areas of debate were left unexplored. Finally, no external ground-truth source for ideological alignment was used, so validation relied solely on comparisons across models and conditions rather than on independent reference points.

These limitations point directly to future directions. A natural extension is to broaden the set of countries and languages, which would allow for a more comprehensive cross-cultural evaluation. Future work could also compare model outputs against external reference sources such as Wikipedia or curated expert-written texts, which would serve as a more objective anchor for assessing ideological stance. Another direction is the use of richer and better-labeled datasets that span a wider range of ideological topics, making it possible to test model behavior beyond the current domains. It would also be valuable to explore repeated prompting of identical arguments to evaluate intra-model variance—that is, whether the same model gives consistent answers to the same question, or whether subtle stochastic effects cause ideological drift. Finally, it is worth considering whether the location from which an LLM is accessed influences its behavior, for example through region-specific filtering, safety layers, or hidden geolocation conditioning. Such factors could further affect the reproducibility and neutrality of model outputs in different contexts.

Overall, these directions suggest that while the present study highlights systematic patterns of ideological bias, there is considerable room to expand the analysis, validate the findings against external standards, and account for geographic and contextual factors that may shape LLM behavior in practice.

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